

AdaPPI: identification of novel protein functional modules via adaptive graph convolution networks in a protein–protein interaction network

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Abstract

Identifying unknown protein functional modules, such as protein complexes and biological pathways, from protein–protein interaction (PPI) networks, provides biologists with an opportunity to efficiently understand cellular function and organization. Finding complex nonlinear relationships in underlying functional modules may involve a long-chain of PPI and pose great challenges in a PPI network with an unevenly sparse and dense node distribution. To overcome these challenges, we propose AdaPPI, an adaptive convolution graph network in PPI networks to predict protein functional modules. We first suggest an attributed graph node presentation algorithm. It can effectively integrate protein gene ontology attributes and network topology, and adaptively aggregates low- or high-order graph structural information according to the node distribution by considering graph node smoothness. Based on the obtained node representations, core cliques and expansion algorithms are applied to find functional modules in PPI networks. Comprehensive performance evaluations and case studies indicate that the framework significantly outperforms state-of-the-art methods. We also presented potential functional modules based on their confidence.

Keywords: protein functional module identification, adaptive graph convolution networks, protein complex, pathway

Introduction

The exploration of biological processes in protein–protein interaction (PPI) networks has contributed to a wide range of functional proteomics studies. This is because biological functions are usually assembled by a group of proteins through interactions with each other. Identifying unknown protein functional modules, such as protein complexes and biological pathways, from PPI networks provide biologists with an opportunity to efficiently understand cellular organization and function. Protein complexes play important roles in biological systems and perform numerous functions such as DNA transcription, mRNA translation and signal transduction [1]. Researchers find new drugs and infer drug–disease relationships through protein complexes [2, 3]. A biological pathway is a series of actions among molecules in a cell that leads to a certain product or change in the cell. The mitogen-activated protein kinase (MAPK), a signaling pathway, is activated by physiological and environmental signals to guide gene expression, leading to cell differentiation, proliferation, apoptosis,

and survival [4]. The identification of pathways facilitates the discovery of a completely unknown pathway or the reconstruction of incomplete pathways, and thereafter helps to understand the process of biochemical reactions within cells [5], such as how drugs are absorbed, distributed, metabolized and excreted in drug discovery [6].

Owing to expensive physicochemical experiments for identifying protein functional modules, such as protein complexes and pathways, researchers have attempted to exploit computational methods to improve its performance by combining various biological information with PPI networks. Protein functional module studies that involve identifying a group of multiple indirectly linked protein entities have been deficient in recent years, which is in sharp contrast to most researches that focus on identifying whether a pair of proteins interact [7–9]. The two tasks are completely different in that functional module identification involves a number of complex combinations of PPI interactions and may cover a longer chain of protein interactions. Without

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candidate protein complex prediction, AF2Complex [10], which tries to predict the structures of multimeric protein complexes would be extremely time-consuming to dock all the possible protein combinations. Furthermore, our approach is the first study to discover a completely novel pathway via an unsupervised clustering approach. Compared with the existing path-finding based methods [11], AdaPPI can be used in the complicated metabolic environment, such as microbial metabolism, with no need of knowing the intermediate or final products of a pathway in priority.

Functional module identification should be modeled as a problem of identifying a group of protein nodes according to their relationships, instead of performing a sequence of link prediction. The relationships between protein interactions in PPI networks can be characterized by an edge in an undirected graph network, in which each node represents a protein with complementary biological information. Two challenges should be addressed in identifying a functional module. First, a good model needs to effectively integrate the node and network topological information. Second, in a PPI network, the whole graph structure will be of unevenly sparse and dense node distribution, and protein nodes with different density distributions in a subgraph must learn node presentations in various neighborhood conditions. A fixed shallow convolution for all nodes in a graph could cause undersmoothing, but a fixed deep convolution creates oversmoothing problems that make the nodes less distinguishable. Learning the protein node information according to the graph node distribution poses another challenge in identifying complex nonlinear relationships in a long chain of protein interactions.

Existing protein functional module identification methods can be roughly divided into two categories.

PPI network structure-based approaches

In this group, only the PPI network was used to measure the relationship between the protein nodes. To predict the intersecting protein complexes, the overlapping clustering method, ClutserOne, finds multiple dense protein groups by adding or removing protein nodes in weighted PPI networks [12]. A group framework [13] obtains all the clusters in PPI networks through multiple clustering methods such as repeated bisections, direct k-way partitioning and multiple k-way partitioning, and then merges clusters with tighter protein node intersections to obtain protein complex sets after cluster purification.

A **clique** is a subset of vertices of an undirected graph in which every pair of distinct vertices is adjacent. High-density cliques are suitable foundation structures for protein complexes [14, 15]. Clustering-based on Maximal Cliques (CMC) [16] calculates the weight of intersections between neighboring nodes, and combines or deletes the cliques according to their intersections in all cliques, and finally predicts protein complexes by filtering the remaining cliques on the weighted sum as density. Considering the chain structure of most protein pathways, the computation of the shallow neighbors of a protein node cannot cover high-order information hidden in PPI networks. Repeated Random Walks (RRW) [17] proposed a biologically sensitive algorithm that used repeated random walks to collect long-range protein interaction information.

Both PPI network structure and biological information based methods

Based on the consideration that only PPI network structure information may be insufficient for measuring similarity between protein nodes, numerous studies have added various biological information about proteins to assist the discovery of functional protein modules. Most proteins can be briefly described in their functions using the existing Gene Ontology (GO)

(<http://geneontology.org/>) attributes including biological processes, molecular functions, and cellular components. The protein complex identification algorithm (PCIA) [18] calculates edge weights using protein gene ontology attributes in PPI networks, chooses protein nodes with high similarity, and then uses the depth-first search to collect the protein complex sets. According to the gene co-expression principle, Xiao et al. [19] builds PPI networks and then detects high-connectivity sub-graphs as cores of protein complexes.

Recently, attributed network embedding based methods, which combine the non-Euclidean PPI network structure and protein biological attributes into a graph and project the graph nodes into a low-dimension space to downstream tasks, have drawn great popularity. Based on the Accelerated Attributed Network Embedding (AANE) model that distributively decomposes the attribute affinity matrix, GO Attributed Network Embedding (GANE) [14] aggregates the protein gene ontology attributes with one-order neighbor information, [20] mines density-based cliques as cores by protein node similarity and expands the cores to protein complexes. Text-associated DeepWalk (TADW) learns graph embedding by incorporating the abundant text content of the nodes, and discovers their maximum commonality with DeepWalk [21, 22]. Based on TADW, the model named Text-associated DeepWalk-spectral Clustering (TADW-SC) utilizes the gene ontology properties of proteins to learn their node representations, thereby obtaining an affinity matrix of protein pairs for spectral clustering [23], and then directly groups protein nodes into clusters as protein complexes [24]. However, both models learned protein node representations using shallow graph neural networks, which may be insufficient for identifying protein complexes or pathways that may require higher-order neighborhood information with a sparse node distribution. The model deepSignalingLinkNet [25] combines the gene expression features of proteins with 3-hop subgraph information while simultaneously learning another genomic representation, by multilayer networks, and concatenates them to classify whether a link in the signaling pathway is true. However, it cannot automatically discover proteins in a pathway. All these methods apply fixed and relatively shallow (low-order) graph convolutions.

The researches suggest that considering the graph diversity will improve the clustering performance of graph neural networks [26, 27]. By learning residual graphs from the data which do not naturally have graph structure and adding them to intrinsic graph structures, Li et al. proposed a graph convolutional networks (GCN) -based model to learn adaptive graph topology structure according to the data and the context of learning task [26]. Considering the diversity of graphs, adaptive graph convolution (AGC) [27] adaptively selects the appropriate order for different graphs instead of adopting the fixed shallow graph convolutions for any graph. The k-order is automatically customized for a graph. However, to identify complex nonlinear relationships in a long chain of protein interaction for functional modules, we should consider the distribution at a node level instead of a graph level to avoid undersmoothing and oversmoothing.

To overcome the fore-mentioned challenges, we proposed AdaPPI, an adaptive framework that can effectively predict protein functional modules: protein complexes and pathways involving a long chain of protein interactions. We first introduced Go Slim information as protein attributes, including 226 unique biological processes and 87 molecular function descriptions. We further proposed an attributed graph node presentation algorithm based on adaptively customized node-wise graph convolution in PPI networks. The algorithm adaptively aggregates low- or high-order graph structure information with protein

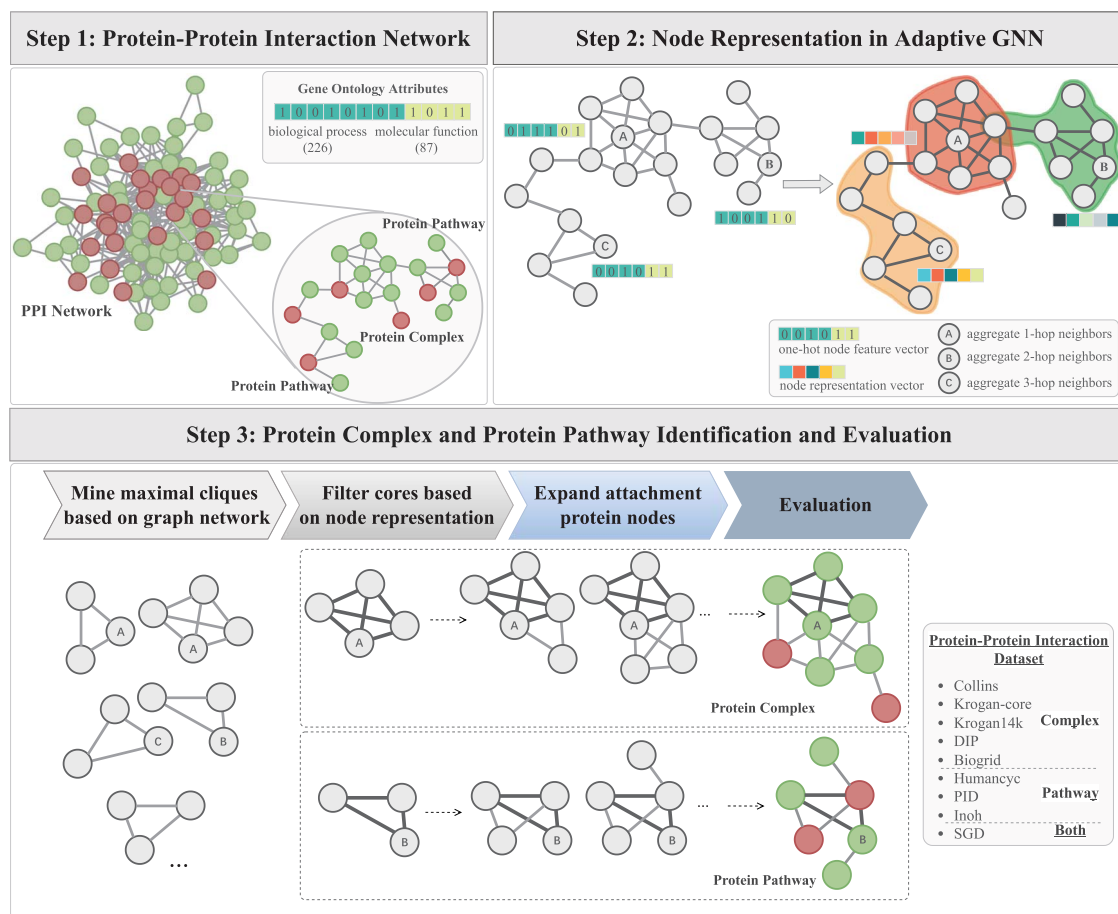


Figure 1. The whole workflow of proposed AdaPPI. Step 1 presents a PPI network including protein complexes and protein pathways, in which each protein node has gene ontology slim attributes that contain biological processes and molecular functions. Step 2 is to learn proteins node representation by adaptively aggregating protein node attributes and various neighborhood structure information in the adaptive graph convolution networks. Step 3 filters cores based on protein node pair similarity after mining all maximal cliques in the graph network, and expands multiple attachment protein nodes to identify protein complexes and protein pathways.

gene ontology attributes to learn low-dimensional protein representations. We performed deep convolution for a node with sparse neighbors, but early automatically terminated convolution when a node had denser neighbors. Thus, the algorithm could capture the information from a long-chain protein interactions in an uneven graph distribution without undersmoothing and oversmoothing. Thereafter, we obtained better node presentations for the PPI network. Third, by mining all cliques in PPI networks we sorted and filtered core clique structures with similarity between protein pairs, based on the node representation calculated in the second step. Finally, we expanded the core cliques with high-correlation attachment protein nodes to obtain a functional protein module, protein complex or pathway. Comprehensive performance evaluations indicate that the framework significantly outperformed state-of-the-art methods. We also presented potential candidate functional modules based on their confidence. The entire framework is shown in Figure 1.

Methodology

PPI interaction network and complementary biological information

PPI network Considering the availability and scale of the datasets, nine PPI datasets were chosen to evaluate algorithm performance, including Collins [28], Krogan-core [29], Krogan14k [29], Biogrid [30], the Database of Interacting Proteins (DIP) [31], Humancyc

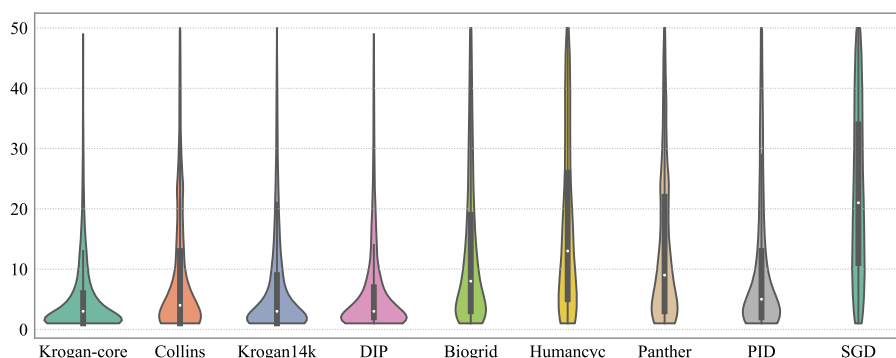
[32], Panthers [33], the Pathway Interaction Database (PID) [34] and the Saccharomyces Genome Database (SGD) [35]. To quantitatively analyze complex and path identification tasks, we also collected the corresponding reference gold-standard functional modules that had more than three proteins from multiple public biological databases: the Munich Information Center for Protein Sequences (MIPS) [36], CYC2008 [37], Aloy [38], SGD [35], TAP06 [39], and Pathway Commons [40]. Table 1 presents the basic statistics of PPI networks. The six PPI networks included a total of 789 reference protein complexes, and three datasets had reference protein pathways of 157, 125, and 217, respectively, whereas SGD network had 104 reference protein pathways and 368 protein complexes.

The violin plots in Figure 2 depict the distribution of all the networks. To illustrate the plots clearly, we trimmed off values larger than 50. The plot shows that SGD has a more elongated distribution, with a more distinct trimmed peak compared with the other data. The density distribution of the data is obviously uneven, where some nodes can learn sufficient information from their direct neighbors while others may need to capture sufficient information from a longer chain of protein interactions.

Biological attributes Each protein in the PPI network can be described using different GO terms: biological process, molecular function and cellular component. Biological processes describe the processes, such as signal transduction and DNA repair, which are vital to living organisms. Molecular function presents

Table 1. Basic statistics of PPI datasets

Type	Dataset	Protein nodes	Interaction edges	GO attributes	Average degree	Max degree
Only	Collins	1622	9074	141	11.189	127
Complex	Krogan-core	2708	7123	143	5.261	141
	Krogan14k	3581	14 076	143	7.861	213
	DIP	4928	17 201	143	6.981	283
	Biogrid	5640	59 748	143	21.187	2570
Only	Humancyc	1648	53 122	151	32.234	366
Pathway	Panther	1802	36 956	142	20.508	234
	PID	2376	39 772	141	16.739	226
	SGD	5808	454 396	181	78.326	2585

**Figure 2.** Degree distributions of all PPI networks with a trim of the upper value 50. The number of nodes with degree greater than 50 for different PPI networks is Collins (67), Krogan-core (9), Krogan14k (57), DIP (51), Biogrid (520), Humancyc (338), Panther (189), PID (194) and SGD (2560), respectively.

molecular-level activity of a single gene product that includes protein, and nucleic acids (RNA), etc. Both biological processes and molecular functions complement PPIs, but without leaking protein complex or pathway information. We excluded cellular component attributes, because the location of gene product activity, such as mitochondria and ribosomes, may also imply complex protein information. The nine datasets included 226 unique biological processes and 87 molecular functions, which were one-hot encoded into 181-dimensional attribute features according to the maximum dimension involved in each PPI network, as shown in Table 1.

Node representations in adaptive graph convolution networks

The node density distribution in a graph is typically uneven. The identification of protein functional modules may involve a long chain of interactions in PPI, which makes it indispensable to explore the global distribution of the graph. To better represent nodes in a graph with unevenly sparse and dense node distributions, graph neural networks should adaptively perform a shallow convolution for a node with dense neighbors, and a deep high-order convolution for a sparse node to capture sufficient information from farther neighbors. A fixed shallow convolution for all nodes in a graph could cause undersmoothing, but a fixed deep convolution creates oversmoothing problems that make the nodes less distinguishable. Therefore, in this study, we evaluated the smoothness of each node and executed an adaptive graph convolution for the graph to perform better node presentations.

Problem definition

Given all protein attributes and the PPI network of an organism, our goal was to effectively integrate the node attributes and network topology to find the appropriate low-dimensional presentations for each node in this network, such that these presentations

can reflect the original semantic relationships between each pair of nodes in this network.

Attributed graph convolution networks

In an attributed graph, the attributes represent the features of a node, whereas structural information indicates the underlying semantic relationship among graph nodes. GCNs-based methods have been proven to be able to naturally integrate the attribute information of nodes and the relationships among nodes [41, 42]. We introduce the input feature matrix $X \in \mathbb{R}^{n \times m}$ for n protein nodes with m attributes, and an adjacency matrix A for the network. The graph Laplacian can be eigen-decomposed as $U\Lambda U^{-1}$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ are the eigenvalues in increasing order and U is the corresponding eigenvector.

By introducing a specified graph filter G and treating m -dimensional data points as m graph signals, a k -order graph convolution can be applied to node X_i as follows:

$$h_i^k = G^k X_i = U p(\Lambda)^k U^{-1} X_i \quad (1)$$

where h_i^k is the current presentation of node i with k -order graph convolution, and $p(\Lambda) = \text{diag}(p(\lambda_1), \dots, p(\lambda_n))$ is called the frequency response function of G .

Adaptive graph convolution networks

This section introduces the smoothness of a graph and evaluates the smoothness saturation of a node. The basic idea of an adaptive graph convolution network is to perform deeper convolution until the specified smoothness saturation of a node arrives, as illustrated in Figure 3. The saturation of node A is satisfied in the 1-hop convolution while the convolution of node C is broken in the third hop.

Smoothness. is an indicator for assessing the degree of similarity of the feature representations among nearby nodes in

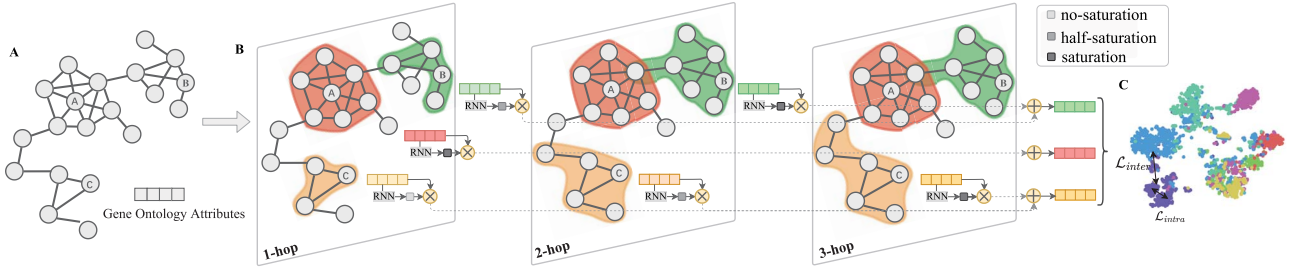


Figure 3. The framework of adaptive graph convolution networks.

a graph, which can be measured using the eigenvalues λ . To smooth the graph, we preserved low-frequency signals, removed high-frequency signals, and assigned $p(\lambda_i)$ with $1 - \frac{1}{2}\lambda_i$ as the study [27].

Thereafter, the smoothness saturation of node i was evaluated as

$$q_i^k = \sigma(w_q \cdot f_0(h_i^k) + b_q) \quad (2)$$

where f_0 is a neural network such as recurrent neural network (RNN). If the accumulated saturation was lower than the smoothness threshold ϵ , namely, $\sum_k q_i^k \leq \epsilon$, more convolutions were executed before reaching the limit of the specified maximum order as follows:

$$h^{k+1} = Gh^k \quad (3)$$

Formulas (2) and (3) customize the k -order convolution for each node. To prevent oversmoothing, once smoothness was saturated the algorithm adaptively terminated the current convolution and recorded the final node embedding as

$$\bar{x}_i = \sum_{k=1}^K q_i^k h_i^k \quad (4)$$

Loss function

To preserve the original semantics of the nodes, we clustered the nodes that were close to each other with similar semantics in PPI into a group. Benefiting from the adaptive convolution termination, the algorithm can effectively prevent the oversmoothing of graph convolution and thereafter make the presentations of nodes in different clusters as distinct as possible. We introduce two loss functions, intracluster and intercluster, and balance them with the parameter β to train the adaptive convolution networks as

$$\mathcal{L} = \mathcal{L}_{intra} + \beta \frac{1}{\mathcal{L}_{inter}} \quad (5)$$

where $\mathcal{L}_{intra}$ makes the nodes in the same cluster as close as possible, defined as

$$\mathcal{L}_{intra} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|c|(|c| - 1)} \sum_{i, j \in c, i \neq j} \|\bar{x}_i - \bar{x}_j\|_2 \quad (6)$$

and $\mathcal{L}_{inter}$ makes the nodes in different clusters as distinctable as possible, which is specified as

$$\mathcal{L}_{inter} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|c|(n - |c|)} \sum_{i \in c, j \notin c} \|\bar{x}_i - \bar{x}_j\|_2 \quad (7)$$

where $|c|$ represents the number of nodes in cluster c , and $|\mathcal{C}|$ denotes the number of clusters.

We then obtained the node presentations of a PPI network, avoiding the oversmoothing and undersmoothing problems. The whole algorithm is presented in Algorithm 1.

Algorithm 1 Node representations in Adaptive Graph Convolution Networks

Input: PPI graph adjacency matrix A , protein node feature matrix X , maximal order of graph convolution K , and smoothness threshold ϵ .

Output: protein node embedding $\bar{X}$.

- 1: Calculate graph filter $G = U(I - \frac{1}{2}\Lambda)U^{-1}$ by A ;
- 2: Initialize a vector s for the sum of smoothness saturation of protein nodes with different orders;
- 3: Initialize a matrix $\bar{X}$ to collect protein node embedding with different orders;
- 4: //Select adaptively different graph convolution layers for the protein nodes
- 5: **for** each $k \in [1, K]$ **do**
- 6: Calculate protein nodes representation h^k by G and X in Equation (1);
- 7: Calculate smoothness saturation q^k by h^k in Equation (7);
- 8: **for** each protein node i **do**
- 9: **if** $s_i \leq \epsilon$ **then**
- 10: Accumulate k -order smoothness $s_i \leftarrow s_i + q_i^k$;
- 11: Update $\bar{x}_i$ by q_i^k and h_i^k in Equation (4);
- 12: **end if**
- 13: **end for**
- 14: **if** All protein nodes finish k -order graph convolution selection **then**
- 15: Break the loop;
- 16: **end if**
- 17: **end for**
- 18: **return** $\bar{X}$

Identification of functional modules based on core-attachment

High-density cliques are more suitable as foundation structures for protein functional modules than clustering [15, 43–45]. A maximum clique is a complete subgraph that is not contained in any other complete subgraph. By default, we refer to a clique as a maximal clique in the rest of the paper. Based on the node embedding obtained, core cliques and expansion algorithms are adopted to identify the functional modules in PPI networks. Similar to GANE [14], we execute core attachment-based module identification in three steps: (1) mining all cliques in the PPI network and sorting

the cliques by density, (2) filtering core structures from the cliques and (3) expanding the cores with attachment nodes.

We first determine all the maximum cliques in an undirected graph by utilizing the depth-first search algorithm proposed in [46]. We then sort the cliques by density computed with node similarity, which assumes that nodes with higher similarity connect tightly. The similarity matrix W of the protein node pairs is obtained by computing the cosine similarity between the protein node embeddings trained by the aforementioned adaptive convolution networks. The density d_s of a clique is evaluated as follows:

$$\begin{cases} w_{ij} = \cos(\bar{x}_i \cdot \bar{x}_j) \\ d_s = \frac{\sum_{v_i, v_j \in c_q} w_{ij}}{|c_q|(|c_q|-1)} \end{cases} \quad (8)$$

where $\bar{x}_i$ denotes the node embedding for node i . $|c_q|$ represents the number of nodes in clique c_q . w_{ij} is the similarity between node v_i and node v_j belonging to the clique c_q . We then sorted and obtained a set of candidate cliques in descending order of density.

In the second step, a series of filtering strategies were designed to obtain the core structures. Cliques whose densities were less than the average density in the candidate set were excluded. In addition, because basic protein nodes can express a variety of functions in multiple complexes or pathways, it is common for a single node to be involved in several different functional modules. We set the clique of the maximum density in the candidate set as the comparison clique and remove it into the core set. We iterate the candidate set. If the current clique has intersection with the comparison clique and also has more than three different nodes simultaneously, we will consider that the clique may belong to a different functional module. However, cliques with fewer than three different nodes should be deleted from the candidate clique set because we assume that the different nodes can not help to form a novel module but only worsen the clique density. We preserved all cliques that had no interaction with the comparison clique.

Finally, iterating all the protein nodes in the PPI network, we identified the attachment nodes of the functional modules, depending on the *correlation* between the new attachment nodes and core cliques.

$$\text{correlation} = \frac{\sum_{v_n, v_j \in c_q} w_{kj}}{|c_q|} \quad (9)$$

where v_n is a new protein node with a core clique c_q from C_c . All the nodes whose *correlation* is larger than the correlation threshold ε are considered qualified attachment nodes to this core according to [14]. In this work, we set it to 0.3, the same as GANE. The entire algorithm is presented in Algorithm 2.

Model Evaluation

Baselines

To examine how biological information is complementary to graph structure in the task of identifying functional modules, we split the baselines into two groups. Considering that finding complex nonlinear relationships in underlying functional modules may involve various steps of protein-protein interactions, we select shallow [14, 24, 47], deep [48] convolution-based methods and the state-of-the-arts adaptive GCN methods [27].

Algorithm 2 Core-Attachment Method of Identification of Complexes and Pathways

Input: mined all cliques set C_s , similarity matrix W , all protein nodes N_s , the correlation threshold ε ,

Output: predicted complexes and pathways set P .

```

1: Calculate all cliques' density scores in  $D_s$  set by  $W$  in equation (8);
2: Sort all cliques set  $C_s$  by their density score  $D_s$  in the descending order;
3: for each clique  $c_q \in C_s$  do
4:   Compare the density  $d_s$  of  $c_q$  with  $\overline{d_s}$  to obtain the candidate set
5: end for
6: // Filter  $C_{cand}$  to find the core cliques set  $C_c$ 
7: while  $|C_{cand}| > 0$  do
8:    $c_m \leftarrow C_{cand}[0]$ ;
9:   Add  $c_m$  to  $C_c$ ;
10:  // Compare the clique with the maximal density score  $c_m$  with each of the remaining cliques  $c_q$ 
11:  for each clique  $c_q \in C_{cand}$  do
12:    Initialize a temp set  $C_{tmp}$ ;
13:    if  $|c_q \cap c_m| == 0$  then
14:      Add  $c_q$  to  $C_{tmp}$ ;
15:    end if
16:    if  $|c_q \cap c_m| \geq 2$  And  $|c_q - c_m| \geq 3$  then
17:      Add  $c_q - c_m$  to  $C_{tmp}$ ;
18:    end if
19:  end for
20:   $C_{cand} \leftarrow C_{tmp}$ ;
21: end while
22: // Iteratively expand attachment protein nodes
23: for each clique  $c_q \in C_c$  do
24:   for a new node  $v_n \in N_s - c_q$  do
25:     Calculate correlation after adding  $v_n$  to  $c_q$  in equation (9);
26:     if correlation  $\geq \varepsilon$  then
27:       Reserve  $v_n$  in  $c_q$ ;
28:     else
29:       Delete  $v_n$  in  $c_q$ ;
30:     end if
31:   end for
32:   Add  $c_q$  to  $P$ ;
33: end for
34: return  $P$ 

```

(i) **Graph structure-based methods.** The Molecular Complex Detection (MCODE) [49], the Algorithm Markov Clustering (MCL) [13], ClusterOne [12], Clustering-based on Maximal Cliques (CMC) [16] measure node similarity only based on the intersection of neighbors between nodes in the PPI network, and then identify the protein complex sets using clusters or cliques. The Protein Complexes from Coherent Partition (PC2P) [50] casts the problem of protein complex prediction as a network partitioning with removal of minimum number of edges.

(ii) **Graph structure and biological information-based methods.** The core-attachment based method (COACH) [15] discovers the preliminary cores based on the average degree of a graph, and then expands the attachment protein nodes with the complementary gene-expression correlations or functional similarities to obtain the high-density protein complex sets. GANE [14] and TADW-SC [24] introduce protein GO information, and integrate graph structure and GO

attributes to learn protein node representations through shallow graph neural networks, predicting protein complex sets based on the core-attachment structure method and spectral clustering, respectively. DPCMNE detects protein complexes via multi-level network embedding [48], learning multi-level network embedding via 40-hop DeepWalk in the Hierarchical Sampling Representation Learning (HSRL) [51], adopting a core-attachment based strategy to detect protein complexes in the weighted PPI network. AGC [27] is a novel low-pass filter based graph convolution algorithms which supports high-order graph convolution but without considering the density distribution difference in a graph. We extend AGC by using the same core-attachment structure method on its node presentations as our proposed method AdaPPI, and name the approach as Protein-AGC. Compared with [27], for AdaPPI, we instead adopt a node-wise adaptive policy, calculate the smoothness saturation of each node and assign different k-order for different nodes, as shown in Formula (2) and (3). R-GMM-VGAE [47] reformulates the GAE-based clustering methodology by proposing a sampling operator against the noisy clustering assignments and an operator against feature drift, which obtains the best performance among the models for attributed graph clustering. We also extend R-GMM-VGAE as what we have dealt with AGC.

Evaluation metrics

We employ the same measures as [52] to explicitly evaluate the performance, including precision, recall, f-score and accuracy.

The number of matched sets between all predicted sets P and all gold-standard sets B determines the algorithm performance. Neighborhood affinity score $NA(p, b)$ is calculated between a predicted protein functional module $p \in P$ and a known standard reference module $b \in B$ as follows:

$$NA(p, b) = \frac{|\mathcal{V}_p \cap \mathcal{V}_b|^2}{|\mathcal{V}_p| \times |\mathcal{V}_b|} \quad (10)$$

where $\mathcal{V}_p$ and $\mathcal{V}_b$ denote the protein vertices of the modules p and b , respectively; $|\cdot|$ is the vertex number of the module. If $NA(p, b)$ is greater than $\varphi = 0.25$, the prediction of the module is considered as a successful match, according to the previous study [16].

Because a predicted protein module may match more than one reference module and vice versa, we count N_{cp} as the number of predicted modules that have at least one reference module and N_{cb} as the number of reference modules that have more than one matched prediction.

Precision was calculated as the proportion of the number of matched predicted modules to the total number of predicted modules:

$$\begin{cases} N_{cp} = |\{p|p \in P, \exists b \in B, NA(p, b) \geq \varphi\}| \\ Precision = \frac{N_{cp}}{|P|} \end{cases} \quad (11)$$

Recall was the proportion of the number of matched reference modules to the number of reference ones.

$$\begin{cases} N_{cb} = |\{b|b \in B, \exists p \in P, NA(p, b) \geq \varphi\}| \\ Recall = \frac{N_{cb}}{|B|} \end{cases} \quad (12)$$

F-score was accordingly determined by

$$F\text{-score} = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (13)$$

The three aforementioned metrics were computed based on the number of matched modules. Another evaluation metric, accuracy, stressed the number of matched proteins included in a module, namely, the clustering-wise accuracy (Acc). T_{ij} represented the number of proteins discovered in the reference module i and predicted module j and N_i denoted the number of proteins in the reference module i . Different from the other three matrices, Acc does not set a threshold for a module using the matched protein proportion.

Acc was calculated as follows:

$$Acc = \sqrt{Sn \times PPV} \quad (14)$$

where the clustering-wise sensitivity (Sn) was determined by

$$Sn = \frac{\sum_{i=1}^{|B|} \max_{j=1}^{|P|} \{T_{ij}\}}{\sum_{i=1}^{|B|} N_i} \quad (15)$$

and the clustering-wise predictive positive value (PPV) was formulated as

$$PPV = \frac{\sum_{j=1}^{|P|} \max_{i=1}^{|B|} \{T_{ij}\}}{\sum_{j=1}^{|P|} \sum_{i=1}^{|B|} T_{ij}} \quad (16)$$

Parameter settings

The parameter settings are described in Table 2. We set the number r of clusters of TADW-SC for the pathway dataset according to the number of reference pathways because TADW-SC was not tested on the pathway dataset: Humancyc, Panther, PID, and SGD. In SGD, we calculate scores with r ranging from 100 to 1000 and select 1000 as the result because TADW-SC obtains a score of zero with other r values. For AdaPPI, we uniformly set the smoothness threshold ϵ to 1. The number r of clusters was estimated using the elbow method based on the sum of the squared errors (SSE) [53] in equation (17):

$$SSE = \sum_{i=1}^r \sum_{v \in C_i} |v - center_i| \quad (17)$$

where $center_i$ is the center point of one cluster C_i generated by the k-means operation in the protein node feature matrix X .

Results and Discussion

Protein complexes identification

In this section, we evaluate the performance of protein complex identification on the Collins, Krogan-core, Krogan14k, Biogrid and DIP datasets. The results are presented in Table 3.

Overall, the proposed AdaPPI achieved the best performance. In all 20 performance scores with respect to the four metrics on the five datasets, the proposed method AdaPPI obtains the 11 (55%) best scores. TADW-SC obtains the four best scores, while GANE gets two. We calculate the average performance on five datasets.

Table 2. Parameter settings

Method	Parameter	Collins	Krogan-core	Krogan14k	DIP	Biogrid	Humancyc	Panther	PID	SGD
TADW-SC	number of clusters r	200	300	400	100	100	200	200	100	1000
R-GMM-VGAE	number of clusters r	30	30	30	30	30	50	50	50	100
	loss coefficient beta1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.05
	loss coefficient beta2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.05	0.01
Protein-AGC	number of clusters r	30	30	30	30	30	50	50	50	100
AdaPPI	number of clusters r	30	30	30	30	30	50	50	50	100
	max order of convolution	10	10	10	10	10	10	10	10	10
	loss coefficient β	10	1	2	1	1	0.1	0.1	0.1	1

Table 3. Performance on protein complex identification in five PPI datasets

Methods	Collins				Krogan-core				Krogan14k			
	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc
MCODE	0.847	0.400	0.540	0.254	0.732	0.198	0.311	0.176	0.612	0.112	0.189	0.152
MCL	0.732	0.506	0.598	0.286	0.335	0.414	0.371	0.262	0.299	0.323	0.311	0.233
ClusterOne	0.733	0.511	0.602	0.290	0.364	0.464	0.408	0.273	0.467	0.302	0.366	0.222
CMC	0.680	0.390	0.496	0.255	0.672	0.304	0.419	0.206	0.472	0.440	0.455	0.210
PC2P	0.731	0.662	0.695	0.303	0.403	0.488	0.441	0.251	0.345	0.380	0.362	0.235
COACH	0.749	0.522	0.615	0.280	0.592	0.460	0.518	0.217	0.461	0.465	0.463	0.217
TADW-SC	0.595	0.793	0.679	0.316	0.426	0.492	0.457	0.297	0.300	0.441	0.357	0.272
GANE	0.819	0.491	0.615	0.293	0.774	0.436	0.558	0.229	0.684	0.442	0.537	0.234
DPCMNE	0.821	0.523	0.639	0.270	0.754	0.455	0.567	0.220	0.621	0.463	0.531	0.223
R-GMM-VGAE	0.890	0.494	0.636	0.256	0.770	0.488	0.597	0.236	0.574	0.361	0.443	0.239
Protein-AGC	0.899	0.608	0.725	0.287	0.680	0.485	0.566	0.233	0.559	0.356	0.435	0.208
AdaPPI	0.967	0.586	0.730	0.284	0.834	0.533	0.650	0.239	0.743	0.493	0.592	0.238

Methods	DIP				Biogrid			
	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc
MCODE	0.542	0.118	0.194	0.149	0.276	0.043	0.075	0.181
MCL	0.153	0.172	0.162	0.190	0.228	0.219	0.223	0.249
ClusterOne	0.390	0.343	0.365	0.227	0.393	0.497	0.439	0.316
CMC	0.603	0.394	0.477	0.219	0.157	0.553	0.245	0.265
PC2P	0.260	0.351	0.299	0.242	–	–	–	–
COACH	0.450	0.620	0.521	0.243	0.311	0.657	0.422	0.276
TADW-SC	0.170	0.084	0.113	0.245	0.550	0.292	0.382	0.292
GANE	0.623	0.550	0.584	0.254	0.545	0.664	0.599	0.310
DPCMNE	0.608	0.575	0.591	0.245	0.447	0.653	0.531	0.304
R-GMM-VGAE	0.574	0.361	0.443	0.239	0.415	0.338	0.373	0.275
Protein-AGC	0.382	0.297	0.334	0.227	0.230	0.269	0.248	0.247
AdaPPI	0.675	0.514	0.583	0.259	0.600	0.535	0.566	0.303

Note: The 1st group is graph structure-based; the 2nd group is graph structure and biological information-based. Bold font indicates the best scores and underlined font indicates the second scores. We did not get the result of PC2P on Biogrid in 7 days.

Compared with the eleven state-of-the-art baselines, AdaPPI outperforms the best baseline TADW-SC in terms of precision, recall, and F-score by 35.56%, 11.18 %, 22.66 % and 7.95%, respectively, but drops 1.98 % under the metric of accuracy.

We noted that compared with TADW-SC, AdaPPI performs not well as respect to Acc. One of the important reasons is that Acc only focuses on counting the number of proteins included in successfully matched complexes, without punishing the unmatched predicted complexes. Therefore, Acc is determined only by the fewer successful match sets, which could be advantageous for methods that tend to work well locally without considering the global situation. However, AdaPPI is suitable for handling the global distribution of graphs. We will perform further analysis with more details in the section “SGD test.”

Influence of biological information. By focusing only on the PPI network structure, MCODE in the first group achieved a precision

score of 84% on Collins but only 27% on Biogrid, which has more protein nodes and greater density variation. The other methods in the first group including MCL, ClusterOne, CMC and PC2P had a similar trend, performing worse as the networks become more complicated. These methods can be influenced tremendously and performed unstably because the similarity of protein nodes was measured based only on graph structure information.

By introducing biological information at the core attachment stage, COACH performs steadily. However, compared with the other four graph neural network-based methods, the protein complex recognition rate of COACH has room for improvement. TADW-SC has the best score in accuracy for Collins, Krogan-core and Krogan14k, but performance drops significantly on other metrics and even further deteriorates on the larger datasets DIP and Biogrid. A possible reason could be that TADW-SC strictly makes a node belong to only one cluster. GANE alleviates this

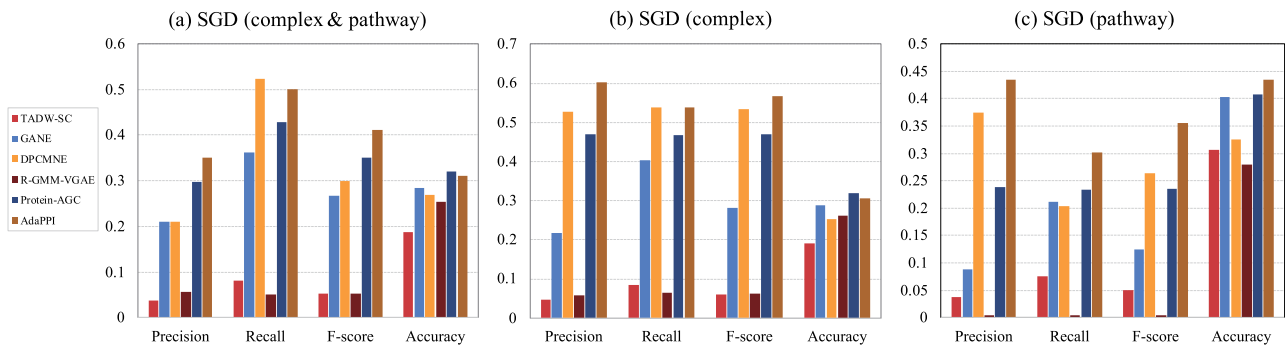
Table 4. Performance on protein pathway identification in three PPI datasets

Methods	Humancyc				Panther				PID			
	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc
TADW-SC	0.130	0.174	0.148	0.387	0.570	0.459	0.509	0.350	0.275	0.304	0.288	0.233
GAINE	0.195	0.146	0.167	0.300	0.481	0.352	0.406	0.272	0.423	0.460	0.441	0.170
DPCMNE	0.191	0.159	0.173	0.292	0.582	0.491	0.533	0.274	0.274	0.314	0.293	0.170
R-GMM-VGAE	0.184	0.128	0.151	0.280	0.684	0.250	0.366	0.251	0.425	0.159	0.232	0.126
Protein-AGC	0.225	0.151	0.181	0.311	0.598	0.431	0.501	0.278	0.511	0.478	0.494	0.180
AdaPPI	0.298	0.174	0.220	0.307	0.660	0.465	0.546	0.273	0.531	0.493	0.511	0.166

Note: The 1st group is graph structure-based; the 2nd group is graph structure and biological information-based. Bold font indicates the best scores and underlined font indicates the second scores.

Table 5. Performance on protein complex and protein identification in SGD dataset

Methods	SGD (complex & pathway)				SGD (only complex)				SGD (only pathway)			
	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc	Precision	Recall	F-score	Acc
TADW-SC	0.038	0.082	0.052	0.188	0.048	0.084	0.061	0.190	0.037	0.076	0.050	0.306
GAINE	0.211	0.362	0.267	0.285	0.217	0.404	0.282	0.287	0.088	0.211	0.124	0.403
DPCMNE	0.210	0.524	0.300	0.269	0.528	0.538	0.533	0.252	0.375	0.203	0.264	0.326
R-GMM-VGAE	0.056	0.051	0.053	0.253	0.059	0.065	0.062	0.261	0.000	0.000	0.000	0.279
Protein-AGC	0.297	0.428	0.351	0.321	0.470	0.467	0.469	0.320	0.238	0.233	0.235	0.408
Protein-NASGC	0.350	0.501	0.412	0.311	0.602	0.538	0.568	0.306	0.434	0.301	0.355	0.434

**Figure 4.** Performance on SGD: (A) Identifies protein complex and protein pathway simultaneously; (B) and (C) identify complexes and pathways, respectively.

problem by setting the cliques to the core structures and then expanding the attachment protein nodes.

Effect of adaptive convolution. The difference between AdaPPI and Protein-AGC lies in the fact that AdaPPI considers the density distribution varying from node to node in a graph and performs adaptive graph convolution accordingly. The results showed that compared with Protein-AGC, AdaPPI achieved a significant improvement. Furthermore, AdaPPI was more robust and stable in all five datasets. Protein-AGC fails to consider the various neighbor situations of each protein node and performs a fixed-order convolution that executes 5, 9, 10, 11 and 13 layers of convolution operations on Collins, Krogan-core, Krogan14k, DIP and Biogrid, respectively. Considering the node distribution, we set the maximum graph convolution order to 10 to effectively mitigate the oversmoothing problem and obtain a more stable performance. As one of the state-of-the-art method for attributed graph clustering, although R-GMM-VGAE obtained the better performance in other tasks [47], its performance was not good in the task of identifying protein complexes. One possible reason is that the task asks for adaptive convolution according to the node distribution of graphs.

Pathway identification

We only compared the performance of the six graph neural network-based methods in the second group because they achieved a significantly good performance. As shown in Table 4, similar to complex identification, the proposed AdaPPI achieves the best performance on the datasets DIP, Humancyc and PID. We will analyze the performance in more detail in the next section because SGD includes both protein complexes and pathways.

SGD test

For SGD test, we compared them under three different conditions: complex-only identification, pathway-only identification, and simultaneous identification of two types of functional modules. We only considered modules with more than three proteins. The predicted modules were matched with the reference complex and pathway. The results are shown in Figure 4. Figure 5 counts the number of predicted modules, matched or unmatched with the references and lists the number of matched reference modules. AdaPPI achieved the best performance, except for the accuracy of complex and simultaneous identification.

Table 6. Identification of L-arginine biosynthesis II (acetyl cycle) pathway

Methods	Pathway protein set of L-arginine biosynthesis II (acetyl cycle)	Number of match
Golden Standard	ARG56, ARG8, ARG2, ARG7, ARG4, ARG1, CPA1, CPA2, URA2, ARG3	10
TADW-SC	XRN1, LEA1, BAT2, COX14, PRO2, TAZ1, ATP7, RDS1, RIP1, CBP4, BI2, CUZ1, COX18, TIF1, SUP35, EMG1, RPS3, RPS19A, REI1, BFR2, CPA1, ARG56	2
GANE	TAZ1, CPA1, CPA2 , HAM1	2
DPCMNE	LOG1, CPA1,CPA2 ,HAM1	2
Protein-AGC	GAP1, ALP1, ORT1, ARG56,ARG3, ARG8, ARG4, ARG1 , CAN1	5
AdaPPI	GAP1, ALP1, SND3, ORT1, ARG56 , IFA38, ARG7, ARG3, ARG8, ARG4 , MST27, ARG1 , CAN1	6

Note: Bold font indicates the matched reference protein's corresponding gene name in the predicted protein pathway.

Effect of adaptive convolution. TADW-SC performs spectral clustering by aggregating 2-hop neighbors information, which groups the majority of nodes into the same cluster in the dense area and forms many small clusters in the sparse area. Therefore, as shown in Figure 5, TADW-SC predicted a large number of modules, with only a few of them matched. That is why a relatively higher Acc is obtained by TADW-SC on all the datasets because Acc is only determined by the fewer successful match sets. GANE, similar to TADW-SC, only captures shallow-order neighbor information but with a core-attachment extension. As shown in Figure 5, 220 predicted modules can be matched with 202 reference complexes and 26 reference pathways, indicating that some predicted modules can be matched with two different modules, which shows that GANE fails to capture the long-chain information and distinguish the characteristics of complexes from pathways. R-GMM-VGAE further deteriorates in the task of identifying pathway modules with the zero score in terms of precision, recall and F-score, which shows the matching proportion is lower than the threshold in Formula (10). The value for Acc is not zero because it does not set a threshold for a module using the matched protein proportion.

Methods	SGD (only complex)		SGD (only pathway)	
	Standard	Matched	Standard	Matched
TADW-SC	368	30	104	8
GANE	368	202	104	26
DPCMNE	368	198	104	15
R-GMM-VGAE	368	32	104	0
Protein-AGC	368	216	104	20
AdaPPI	368	227	104	30

Note: Standard means the number of golden standard sets; Matched means the number of matched modules.

Figure 6 illustrates the selected convolution orders of AdaPPI in the SGD dataset. More than 3000 protein nodes choose to terminate learning after the 2-layer graph convolution operation, and approximately 1000 protein nodes select seven layers. Moreover, compared with Protein-AGC, which aggregates the fixed 7th-order neighbor information, the ratio of the matched complexes to pathways of AdaPPI is closer to the reference ratio of SGC, which demonstrates that AdaPPI captures higher-order pathway information while balancing the complex identification. The proposed adaptive convolution achieves greater improvement by considering node distribution.

We note that although AdaPPI obtained relatively good performance, the scores for all the methods are not very high. As shown

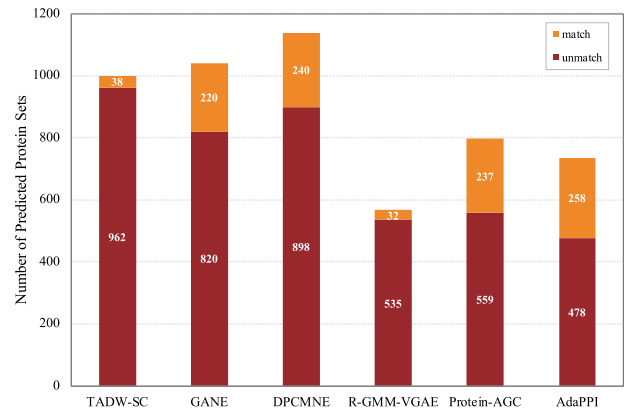


Figure 5. The number of predicted modules: the orange part indicates the matched predicted modules with a complex or a pathway; and the red part shows the number of unmatched predictions.

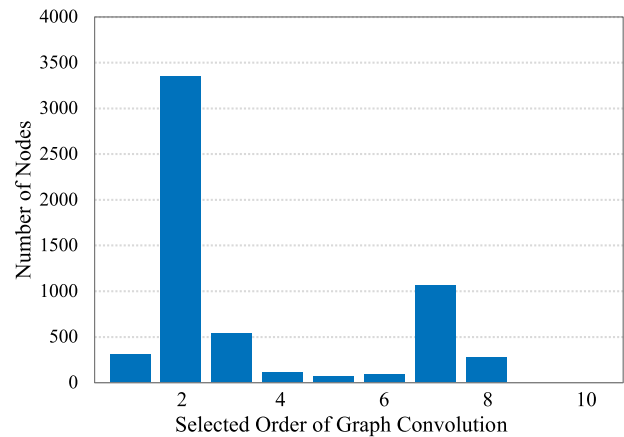


Figure 6. Number of protein nodes with respect to the selected order of graph convolution for the SGD dataset.

in Table 1, SGD covers 5808 protein nodes, 454,396 interactions and 181 Go attributes. Its average node degree is 78.326 and the maximum degree is 2585. The network itself is quite complicated with uneven node degree distribution and complex interaction relationships. The methods get at least 736 clusters. Averagely, eight nodes form a cluster. Clustering such a network is non-trivial due to its intrinsic complexity. Core attachment expansion algorithm further deteriorates the accuracy, because there is no standard value to break the attachment and an empirical value as shown in Formula 9.

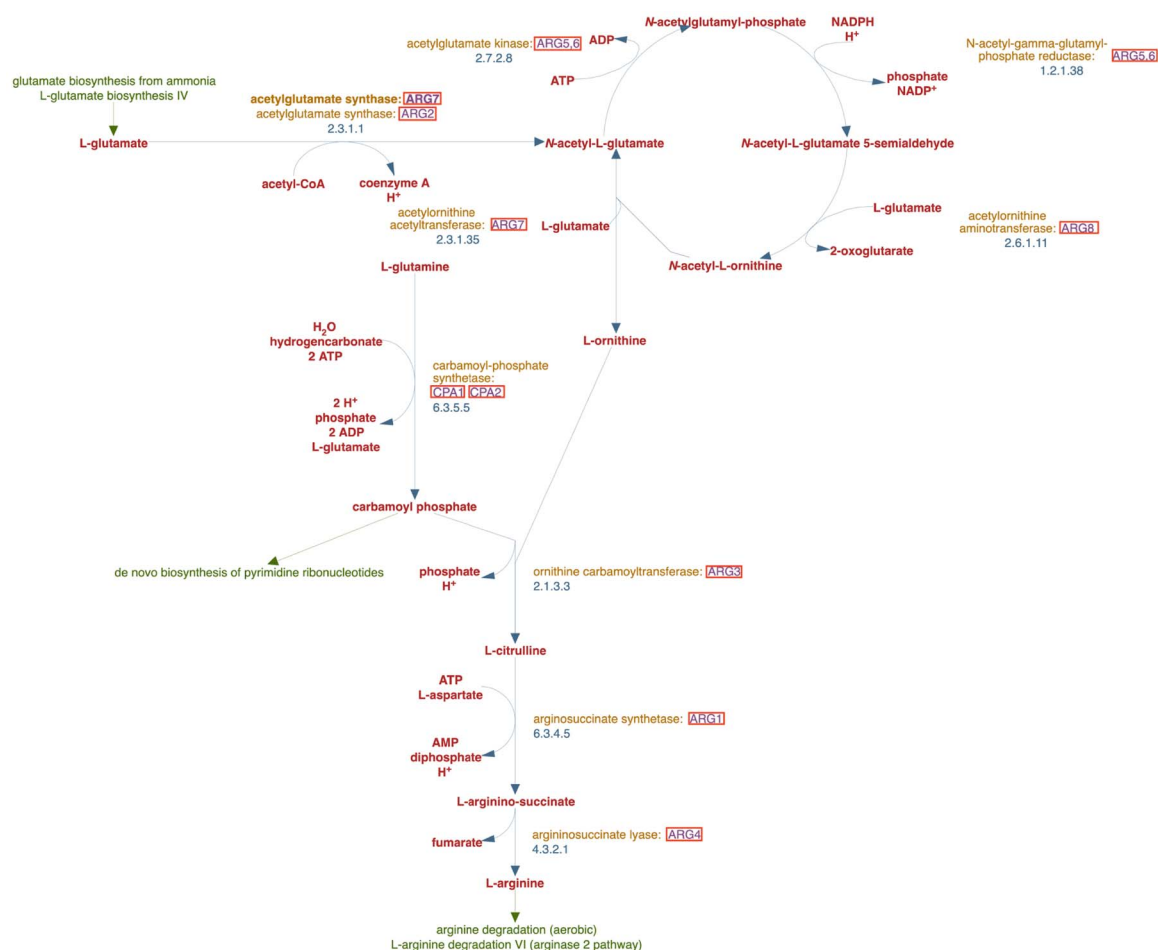


Figure 7. *Saccharomyces cerevisiae* pathway: L-arginine biosynthesis II (acetyl cycle).

Case studies

We further explain the effect of AdaPPI using two typical examples: the L-arginine biosynthesis II (acetyl cycle) pathway and the protein RSC complex.

Pathway identification

This case study used Arginine biosynthesis as an example to show the details of how AdaPPI identifies the pathway with the PPI network. Arginine biosynthesis from YeastPathways (<https://pathway.yeastgenome.org/>) curated by SGD is notable for its complexity and variability at the genetic level due to its connection with several other pathways, such as pyrimidine and polyamine biosynthesis, and certain degradative pathways. As shown in Figure 7, Arginine biosynthesis in *Saccharomyces cerevisiae* occurs in two different locations, the mitochondria and cytoplasm. The biosynthesis of the first five steps in mitochondria is known as acetylated derivative cycle. The acetyl group is added to L-glutamate in the first step via acetylornithine acetyltransferase (ARG2) and acetylglutamate synthase (ARG76). The enzymes that catalyze the second and third steps, N-acetylglutamate kinase and N-acetylglutamyl-phosphate reductase, are encoded by a single gene (ARG5,6) [54]. Ornithine, an intermediate product, is produced in mitochondria and exported to the cytoplasm by a transporter [55]. L-ornithine is then converted into L-arginine in the cytoplasm via three reactions mediated by ornithine carbamoyltransferase (ARG3), arginosuccinate synthase (ARG1), and argininosuccinate lyase (ARG4). In brief, with the proposed AdaPPI,

six of the 10 genes encoding the necessary proteases in biosynthesis – ARG56, ARG7 and ARG8 involved in the reactions in the mitochondria and ARG3, ARG1, ARG4 in the cytoplasm—are successfully identified. Table 7 presents the predicted modules and Figure 8 displays the related interaction networks of different methods. The red connecting nodes imply that these nodes directly connect with each other but do not belong to the module because the pathway covers a long chain of protein interactions.

Protein complex case

The remodeling the structure of chromatin (RSC) complex is a member of ATP-dependent chromatin remodelers that contribute to the integrity of centromeric DNA [56]. We can see that the red nodes are fewer than those in the pathway identification. Because most protein complexes are generated by contact between proteins, and all the found nodes tend to be intermediate neighbors in the network. Therefore, fewer red connecting nodes are needed.

Potential candidate functional modules

This section provides the predicted functional modules, which do not match the existing functional modules, but could be potential candidate functional modules with a certain confidence. The *P*-value refers to the statistical significance of the occurrence of a predicted protein functional module with respect to a given

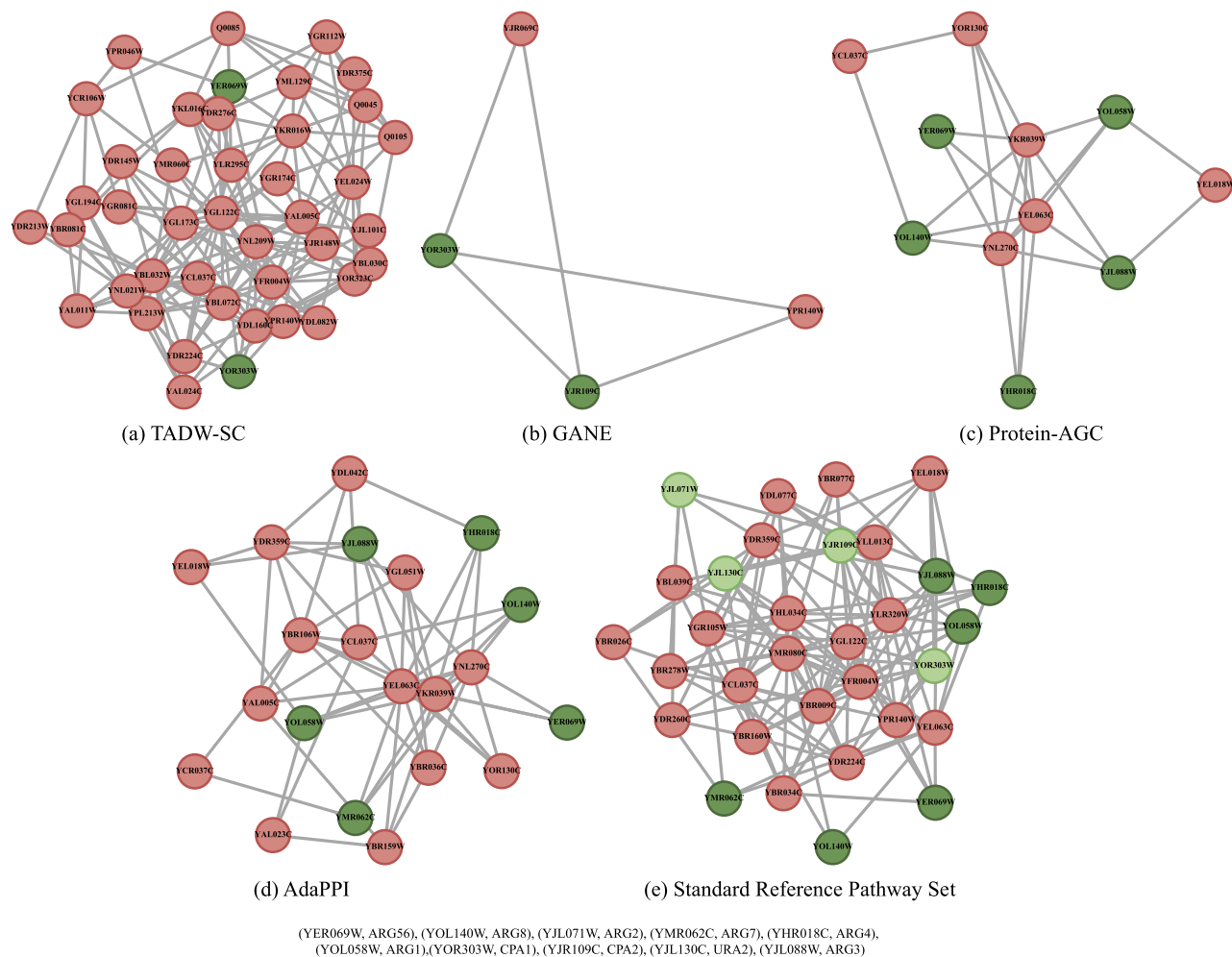


Figure 8. (A)-(D) are subgraphs of identified protein pathway based on four different methods; (E) is a subgraph of the standard protein pathway for L-arginine biosynthesis II (acetyl cycle). The dark green nodes are the matched nodes of AdaPPI, the light green nodes are those which belong to the pathway but fail to be predicted, and the red ones are only used as the connecting nodes in the PPI network. Please note that the subgraphs are manually generated with the shortest path algorithm between node pairs to work as a reference. The names of the basic proteins forming the protein pathway and their corresponding gene name are listed in pairs in bottom.

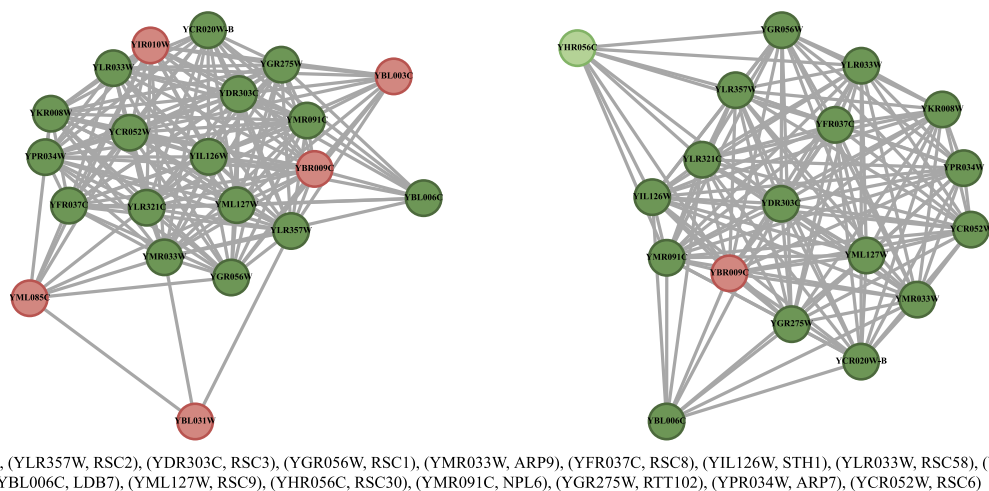


Figure 9. The identified vs. the standard protein complex RSC.

Table 7. Analysis in unmatched samples of predicted protein pathways in SGD

Index	Unmatched samples of predicted protein pathway	Min P-value	GO ID	GO Term
1	YEL002C YEL042W YOR085W YMR149W YER005W YOR067C YBR171W YLR292C YDR245W YPL227C YDR414C YEL036C YNL219C YJL183W YOR002W	3.720e-22	GO:0006486	Protein glycosylation
2	YCR034W YBR026C YHL048W YLR372W YKL055C YOR221C YMR207C YMR272C YJL196C YGR295C	9.842e-17	GO:0006633	Fatty acid biosynthetic process
3	YBL045C YDR322C-A YPR191W YDR529C YGL187C YFR033C YOR065W YJL045W YHR001W-A YGR183C	1.694e-15	GO:0019646	Aerobic electron transport chain
4	YBL064C YBR244W YML028W YDR453C YIR037W YKL026C YCL037C YLR109W	1.117e-14	GO:0098869	Cellular oxidant detoxification
5	Q0250 Q0275 YPL189C-A YGR112W YMR257C YIL157C YJL062W-A YLR203C Q0045 YML129C	1.425e-13	GO:0008535	Respiratory chain complex IV assembly

Table 8. Analysis of unmatched samples of the predicted protein complexes in SGD

Index	Unmatched samples of predicted protein complex	Min P-value	GO ID	GO Term
1	YMR224C YNL218W YHR031C YNL262W YNL273W YPL024W YMR078C YHR191C YBR223C YPL008W YOR144C YCL061C YDR217C YJL092W YGL163C YGL175C YMR048W YHR164C YJL173C YJR043C YAR007C YBL088C YLR288C YDL220C YKL113C YLR320W YER095W YDR359C YBR136W YCR066W YDL059C YPR135W YPL022W YBR073W YBR098W YML032C YMR190C YDR004W YMR284W YNL250W YNL312W YDR180W YOR033C YDR386W YBR088C	7.180e-52	GO:0006281	DNA repair
2	YPL178W YKL173W YDL209C YPR101W YMR240C YMR213W YOR308C YNR011C YFL017W-A YPR178W YIR005W YGR129W YLR016C YPL213W YJR084W YBR055C YLR438C-A YLR298C YCR063W YDR482C YPL064C YBR065C YBR188C YHR086W YER029C YJL203W YGL174W YPR045C YPR182W	1.66e-47	GO:0000398	mRNA splicing, via spliceosome
3	YNL002C YGL076C YBR142W YPL093W YFR001W YKL009W YAL025C YGR103W YGL111W YKL021C YMR049C YNL061W YLR002C YNL110C YOL077C YOR272W YER126C YPL043W YML065W YLR009W YDR060W YKL082C YER002W YMR290C YPL211W YPR016C YOR294W YOR206W	2.178e-45	GO:0042273	Ribosomal large subunit biogenesis
4	YDR365C YCL059C YKR060W YCR057C YDR299W YJL010C YGR145W YER082C YOR310C YPL126W YHL034C YBL004W YJL109C YDR449C YDL148C YNR054C YOR287C YGR090W YHR196W YJR002W YLR409C YIL019W YDR324C YPR144C	2.039e-34	GO:0000462	Maturation of SSU-rRNA from tricistronic rRNA transcript (SSU-rRNA, 5.8S rRNA, LSU-rRNA)
5	YAR007C YBL088C YDL002C YDR369C YKL113C YER092W YER095W YBR136W YDL059C YGR099W YOR033C YLR052W YMR078C YML032C YHR164C YCL061C YMR284W YNL312W YMR190C YOR189W	1.634e-31	GO:0000723	Telomere maintenance

functional annotation, which is computed by the following hypergeometric distribution:

$$P\text{-value} = 1 - \sum_{i=0}^{t-1} \frac{\binom{|F|}{i} \binom{|V|-|F|}{|M|-i}}{\binom{|V|}{|M|}} \quad (17)$$

where the predicted module M contains t proteins in functional group F and the entire PPI network contains $|V|$ proteins. We calculated the P -value using tool GO::TermFinder (<https://www.yeastgenome.org/goSlimMapper>) (<https://go.princeton.edu/cgi-bin/GOTermFinder>) to evaluate the biological sense of the predicted functional modules with a cutoff being 0.01. Tables 8 and 9 provide five top predicted but unmatched protein pathways

and complexes in SGD dataset. More detailed modules can be found in the supplementary materials.

Conclusion

Protein functional modules provide biologists with an opportunity to efficiently understand cellular organizations and functions. However, many functional modules, such as protein complexes and biological pathways, remain unknown. Given a PPI network with unevenly sparse and dense node distribution and all protein attributes, how to effectively integrate the node attributes and network topology to find the appropriate low-dimensional presentations for each node in this network, such that these presentations can reflect the original semantic relationships between

each pair of nodes in this network propose great challenges. We propose an attributed graph node presentation algorithm that can adaptively aggregate low- or high-order graph structure information with protein gene ontology attributes to learn low-dimensional representations based on the predicted graph node smoothness. Based on the obtained node embedding, we propose core cliques and expansion algorithms to find functional modules in PPI networks. Comprehensive performance evaluations indicate that the framework significantly outperforms state-of-the-art methods. We further present the potential candidate functional modules according to their confidence. To further validate the proposed candidate modules in the future, we need to adjust the correlation threshold to make the new attachment nodes and core cliques closer, and use the domain knowledge to select meaningful candidate protein complex and pathway modules.

Key Points

- We propose AdaPPI, a deep learning based framework to predict unknown protein functional modules, such as protein complexes and pathways, from PPI networks.
- Our approach is the first study to discover a completely novel pathway via an unsupervised clustering approach. Compared with the existing path-finding based methods, AdaPPI can be used in the complicated metabolic environment with no need of knowing the intermediate or final products of a pathway in priority.
- Identifying the protein complex modules, AdaPPI can reduce the search space for the tools which tries to predict the structures of multimeric protein complexes, such as AlphaFold2 and AF2Complex.
- Considering the uneven dense and sparse node distribution in a graph, the study proposes a node representation algorithm via adaptive GCN. The comprehensive performance evaluations and case studies indicate that the framework significantly outperforms the state-of-the-art methods.

Supplementary Data

Supplementary data are available online at https://github.com/al-area/AdaPPI/tree/main/supplementary_files.

Data availability

The source code and data are available at <https://github.com/al-area/AdaPPI>.

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References

1. Xu B, Liu Y, Lin C, et al. Reconstruction of the protein-protein interaction network for protein complexes identification by walking on the protein pair fingerprints similarity network. *Front Genet* 2018;**9**:272.
2. Suthram S, Dudley JT, Chiang AP, et al. Network-based elucidation of human disease similarities reveals common functional modules enriched for pluripotent drug targets. *PLoS Comput Biol* 2010;**6**(2):e1000662.
3. Yu L, Huang J, Ma Z, et al. Inferring drug-disease associations based on known protein complexes. *BMC Med Genomics* 2015;**8**(2):1–13.
4. Seger R, Wexler S. The MAPK signaling cascades. *Encyclopedia Cell Biol* 2016;**3**(9):122–7.
5. Abadjieva A, Pauwels K, Hilven P, et al. A new yeast metabolon involving at least the two first enzymes of arginine biosynthesis: acetylglutamate synthase activity requires complex formation with acetylglutamate kinase. *J Biol Chem* 2001;**276**(46):42869–80.
6. Shah HA, Liu J, Yang Z, et al. Review of machine learning methods for the prediction and reconstruction of metabolic pathways. *Front Mol Biosci* 2021;**6**:8.
7. Raimondi D, Simm J, Arany A, et al. A novel method for data fusion over entity-relation graphs and its application to protein-protein interaction prediction. *Bioinformatics* 2021;**37**(16):2275–81.
8. Nasiri E, Berahmand K, Rostami M, et al. A novel link prediction algorithm for protein-protein interaction networks by attributed graph embedding. *Comput Biol Med* 2021;**137**:104772.
9. Hu X, Feng C, Zhou Y, et al. DeepTrio: a ternary prediction system for protein-protein interaction using mask multiple parallel convolutional neural networks. *Bioinformatics* 2022;**38**(3):694–702.
10. Gao M, Nakajima An D, Parks JM, et al. AF2Complex predicts direct physical interactions in multimeric proteins with deep learning. *Nat Commun* 2022;**13**(1):1–13.
11. Bebek G, Yang J. PathFinder: mining signal transduction pathway segments from protein-protein interaction networks. *BMC Bioinformatics* 2007;**8**(1):1–13.
12. Nepusz T, Yu H, Paccanaro A. Detecting overlapping protein complexes in protein-protein interaction networks. *Nat Methods* 2012;**9**(5):471–2.
13. Asur S, Ucar D, Parthasarathy S. An ensemble framework for clustering protein-protein interaction networks. *Bioinformatics* 2007;**23**(13):i29–40.
14. Xu B, Li K, Zheng W, et al. Protein complexes identification based on go attributed network embedding. *BMC Bioinformatics* 2018;**19**(1):1–10.
15. Wu M, Li X, Kwok CK, et al. A core-attachment based method to detect protein complexes in PPI networks. *BMC Bioinformatics* 2009;**10**(1):1–16.
16. Liu G, Wong L, Chua HN. Complex discovery from weighted PPI networks. *Bioinformatics* 2009;**25**(15):1891–7.
17. Macropol K, Can T, Singh AK. RRW: repeated random walks on genome-scale protein networks for local cluster discovery. *BMC Bioinformatics* 2009;**10**(1):1–10.
18. Hu AL, Chan KC. Utilizing both topological and attribute information for protein complex identification in PPI networks. *IEEE/ACM Trans Comput Biol Bioinform* 2013;**10**(3):780–92.
19. Xiao Q, Luo P, Li M, et al. A novel core-attachment-based method to identify dynamic protein complexes based on gene expression profiles and PPI networks. *Proteomics* 2019;**19**(5):1800129.
20. Huang X, Li J, Hu X. Accelerated attributed network embedding. In: *Proceedings of the 2017 SIAM International Conference on Data Mining*. SIAM, 2017, 633–41.
21. Perozzi B, Al-Rfou R, Skiena S. Deepwalk: online learning of social representations. In: *Proceedings of the 20th ACM SIGKDD*

- International Conference on Knowledge Discovery and Data Mining. Association for Computing Machinery, 2014, 701–10.
22. Yang C, Liu Z, Zhao D, et al. Network representation learning with rich text information. In: *Proceedings of the 24th International Joint Conference on Artificial Intelligence*. AAAI Press, 2015, 2111–7.
 23. Von Luxburg U. A tutorial on spectral clustering. *Stat Comput* 2007;**17**(4):395–416.
 24. Berahmand K, Nasiri E, Li Y, et al. Spectral clustering on protein-protein interaction networks via constructing affinity matrix using attributed graph embedding. *Comput Biol Med* 2021;**138**:104933.
 25. Feng J, Zeng A, Chen Y, et al. Signaling interaction link prediction using deep graph neural networks integrating protein-protein interactions and omics data. *BioRxiv* 2020;2020.12.23.424230. <https://doi.org/10.1101/2020.12.23.424230>.
 26. Li R, Wang S, Zhu F, et al. Adaptive graph convolutional neural networks. *Proc AAAI Conf Artif Intell* 2018;**32**(1):8.
 27. Zhang X, Liu H, Li Q, et al. Attributed graph clustering via adaptive graph convolution. In: *Proceedings of the 28th International Joint Conference on Artificial Intelligence*. AAAI Press, 2019, 4327–33.
 28. Collins SR, Kemmeren P, Zhao XC, et al. Toward a comprehensive atlas of the physical interactome of *Saccharomyces cerevisiae*. *Mol Cell Proteomics* 2007;**6**(3):439–50.
 29. Krogan NJ, Cagney G, Yu H, et al. Global landscape of protein complexes in the yeast *Saccharomyces cerevisiae*. *Nature* 2006;**440**(7084):637–43.
 30. Stark C, Breitkreutz BJ, Reguly T, et al. BioGRID: a general repository for interaction datasets. *Nucleic Acids Res* 2006;**34**(suppl_1):D535–9.
 31. Xenarios I, Salwinski L, Duan XJ, et al. DIP, the database of interacting proteins: a research tool for studying cellular networks of protein interactions. *Nucleic Acids Res* 2002;**30**(1):303–5.
 32. Romero P, Wagg J, Green ML, et al. Computational prediction of human metabolic pathways from the complete human genome. *Genome Biol* 2005;**6**(1):1–17.
 33. Mi H, Huang X, Muruganujan A, et al. PANTHER version 11: expanded annotation data from Gene Ontology and Reactome pathways, and data analysis tool enhancements. *Nucleic Acids Res* 2017;**45**(D1):D183–9.
 34. Schaefer CF, Anthony K, Krupa S, et al. PID: the pathway interaction database. *Nucleic Acids Res* 2009;**37**(suppl_1):D674–9.
 35. Cherry JM, Adler C, Ball C, et al. SGD: *Saccharomyces* genome database. *Nucleic Acids Res* 1998;**26**(1):73–9.
 36. Mewes HW, Amid C, Arnold R, et al. MIPS: analysis and annotation of proteins from whole genomes. *Nucleic Acids Res* 2004;**32**(suppl_1):D41–4.
 37. Pu S, Wong J, Turner B, et al. Up-to-date catalogues of yeast protein complexes. *Nucleic Acids Res* 2009;**37**(3):825–31.
 38. Aloy P, Bottcher B, Ceulemans H, et al. Structure-based assembly of protein complexes in yeast. *Science* 2004;**303**(5666):2026–9.
 39. Gavin AC, Aloy P, Grandi P, et al. Proteome survey reveals modularity of the yeast cell machinery. *Nature* 2006;**440**(7084):631–6.
 40. Rodchenkov I, Babur O, Luna A, et al. Pathway commons 2019 update: integration, analysis and exploration of pathway data. *Nucleic Acids Res* 2020;**48**(D1):D489–97.
 41. Kipf TN, Welling M. Semi-supervised classification with graph convolutional networks. In: *Proceedings of the 5th International Conference on Learning Representations*. OpenReview.net, 2017.
 42. Ji C, Chen H, Wang R, et al. Smoothness sensor: adaptive smoothness-transition graph convolutions for attributed graph clustering. *IEEE Trans Cybernetics* 2021;**52**:12771–84.
 43. Leung HC, Xiang Q, Yiu SM, et al. Predicting protein complexes from PPI data: a core-attachment approach. *J Comput Biol* 2009;**16**(2):133–44.
 44. Peng W, Wang J, Zhao B, et al. Identification of protein complexes using weighted Pagerank-nibble algorithm and core-attachment structure. *IEEE/ACM Trans Comput Biol Bioinform* 2015;**09**(12):179–92.
 45. Wang R, Liu G, Wang C. Identifying protein complexes based on an edge weight algorithm and core-attachment structure. *BMC Bioinformatics* 2019;**09**:20.
 46. Bron C, Kerbosch J. Algorithm 457: finding all cliques of an undirected graph. *Commun ACM* 1973;**16**(9):575–7.
 47. Mrabah N, Bouguessa M, Touati MF, et al. Rethinking graph auto-encoder models for attributed graph clustering. *ArXiv* 2022;**abs/2107.08562**.
 48. Meng X, Xiang J, Zheng R, et al. DPCMNE: detecting protein complexes from protein-protein interaction networks via multi-level network embedding. *IEEE/ACM Trans Comput Biol Bioinform* 2021;**19**(3):1592–602.
 49. Bader GD, Hogue CWV. An automated method for finding molecular complexes in large protein interaction networks. *BMC Bioinformatics* 2003;**4**:2.
 50. Omranian S, Angeleska A, Nikoloski Z. PC2P: parameter-free network-based prediction of protein complexes. *Bioinformatics* 2021;**37**(1):73–81.
 51. Fu G, Hou C, Yao X. Learning topological representation for networks via hierarchical sampling. In: *2019 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2019, 1–8.
 52. Zaki N, Efimov D, Berengueres J. Protein complex detection using interaction reliability assessment and weighted clustering coefficient. *BMC Bioinformatics* 2013;**14**(1):1–9.
 53. Thorndike RL. *Who belongs in the family*. Psychometrika. Citeseer, 1953.
 54. Boonchird C, Messenguy F, Dubois E. Determination of amino acid sequences involved in the processing of the ARG5/ARG6 precursor in *Saccharomyces cerevisiae*. *Eur J Biochem* 1991;**199**(2):325–35.
 55. Crabeel M, Soetens O, De Rijcke M, et al. The ARG11 gene of *Saccharomyces cerevisiae* encodes a mitochondrial integral membrane protein required for arginine biosynthesis. *J Biol Chem* 1996;**271**(40):25011–8.
 56. Cairns BR, Lorch Y, Li Y, et al. RSC, an essential, abundant chromatin-remodeling complex. *Cell* 1996;**87**(7):1249–60.